Leonardo Da Pinchy

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***Abstract* — Leonardo Da Pinchy is a six-degree-of-freedom robotic arm for transferring SVG artwork onto known curved surfaces. A Jetson Orin Nano runs our ROS2 pipeline for artwork mapping, AprilTag-based fixture registration, and drawing execution. Six positioning joints orient a manually loaded pen; a separate servo is planned for gripping. The system uses a $0.4 \times 0.4 \times 0.4$ m working region and targets the cylinder's accessible front 180 degrees. Camera registration and pen-tip positioning each have a separate 2 mm error target.**

***Index Terms* — AprilTags, computer vision, inverse kinematics, robot arm, robotic drawing, ROS2, surface mapping**

I. INTRODUCTION

Digital artists often create designs in two dimensions but want to reproduce them on physical objects such as mugs or curved display pieces. Drawing each copy by hand requires translating a flat design onto a curved surface, which becomes tedious for repeated production. This project develops a robot that performs that transfer while preserving the artist's control over the original design, its placement, and its scale.

Our intended user is an independent creator, student, or small business producing custom artwork in small batches. The user supplies a scalable vector graphic (SVG), selects an object model, and places the object on a locating platform. The system then places the drawing on the model, finds where the physical object sits relative to the arm, and guides the pen across its surface while keeping it perpendicular to the surface.

A conventional planar plotter can reproduce line artwork on a flat sheet, but drawing on a curved surface requires control of the tool's pointing direction as well as its position. Our 6 degrees of freedom (DOF) robotic arm adjusts the pen's orientation along the surface, allowing the artist to reproduce digital strokes on supported curved objects. The joint mechanics and lightweight printed links draw inspiration from the SO-101 [1], while the UR3e's six-axis layout informs the positioning configuration [2]. We implement artwork mapping and object registration, using ROS2 and MoveIt 2 to plan and execute arm motion.

Our longer-term goal is to draw on objects whose pose and surface are detected by depth cameras, without AprilTags or a locating platform. The artist would place artwork on the detected surface and submit the job. For the first version, we use a known cylinder and locating platform to test drawing before adding surface reconstruction.

II. USE-CASE REQUIREMENTS

The system must reproduce an artist's SVG line artwork on the accessible front 180-degree region of one known, stationary cylinder or cylindrical mug. The drawing surface must have constant radius and remain clear of handles and fixtures. The object and its holder must fit within the designated $0.4 \times 0.4 \times 0.4$ m working region, with all drawing and pen-up motions reachable. Unknown, tapered, spherical, and concave surfaces are outside the initial scope.

The artist must be able to upload artwork, choose scale and placement, preview the result, and start a job through a browser. Initial operation uses a manually loaded pen and one ink color. Jobs must preserve separate strokes without unintended connecting marks. The provisional supported input is at most 50 strokes and 1 m of mapped drawing path. Automatic pen pickup is a later addition.

To preserve the artist's design, a completed drawing must contain every intended stroke without unintended connecting marks. After aligning a scan with the SVG by translation and rotation only, the maximum distance between corresponding stroke centerlines must be no greater than 2 mm. Scaling or warping the scan is not permitted, since it could hide changes to the drawing's size or shape. Sec. VII.C evaluates this requirement across five platform placements, including a job at the provisional input limits.

The system must reject invalid artwork, unreliable registration, and unreachable or colliding paths before the arm moves, and display the reason. Software Stop must hold the arm; the emergency stop must cut motor power. The shape requirement above is separate from the 2 mm registration and pen-tip targets in Sec. IV. Overall placement error will also be measured before scan alignment; its acceptance limit remains to be finalized.

III. ARCHITECTURE AND/OR PRINCIPLE OF OPERATION

Fig. 1 shows the physical system, and Fig. 2 identifies the main functions and interfaces. The magnified physical view is in Appendix A. Implementation details appear in Sec. VI.

A. External Computer

The browser lets the artist upload an SVG, choose its scale and placement, and preview the drawing before selecting Start. It exchanges artwork, settings, and job status with the Jetson over HTTP, and provides a Stop control.

B. Jetson Orin Nano

The Jetson places the SVG strokes on the known cylinder, then uses the camera and fixture geometry to locate them relative to the arm. Before motion begins, it checks the artwork, the estimated object position, and whether the arm can follow the path smoothly without collisions. It then sends timed joint movements through the driver board and reads the servo feedback. RViz2 shows the robot and planned paths for inspection [4].

C. Robot Arm Hardware

Six servo-driven joints move the manually loaded pen along the curved surface while pegs hold the object on its tagged platform. A dedicated supply powers the motors through the emergency stop, and USB and UART connections carry commands and feedback. Automatic gripping is planned for a later version.

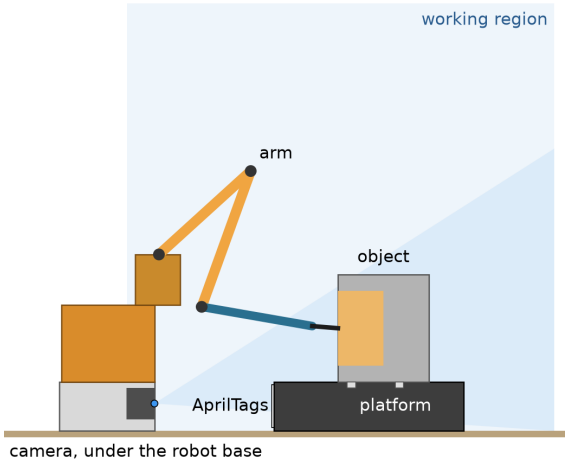


Figure 1: Physical arrangement. The base-mounted D455 observes platform tags; the blue region is the designated working volume.

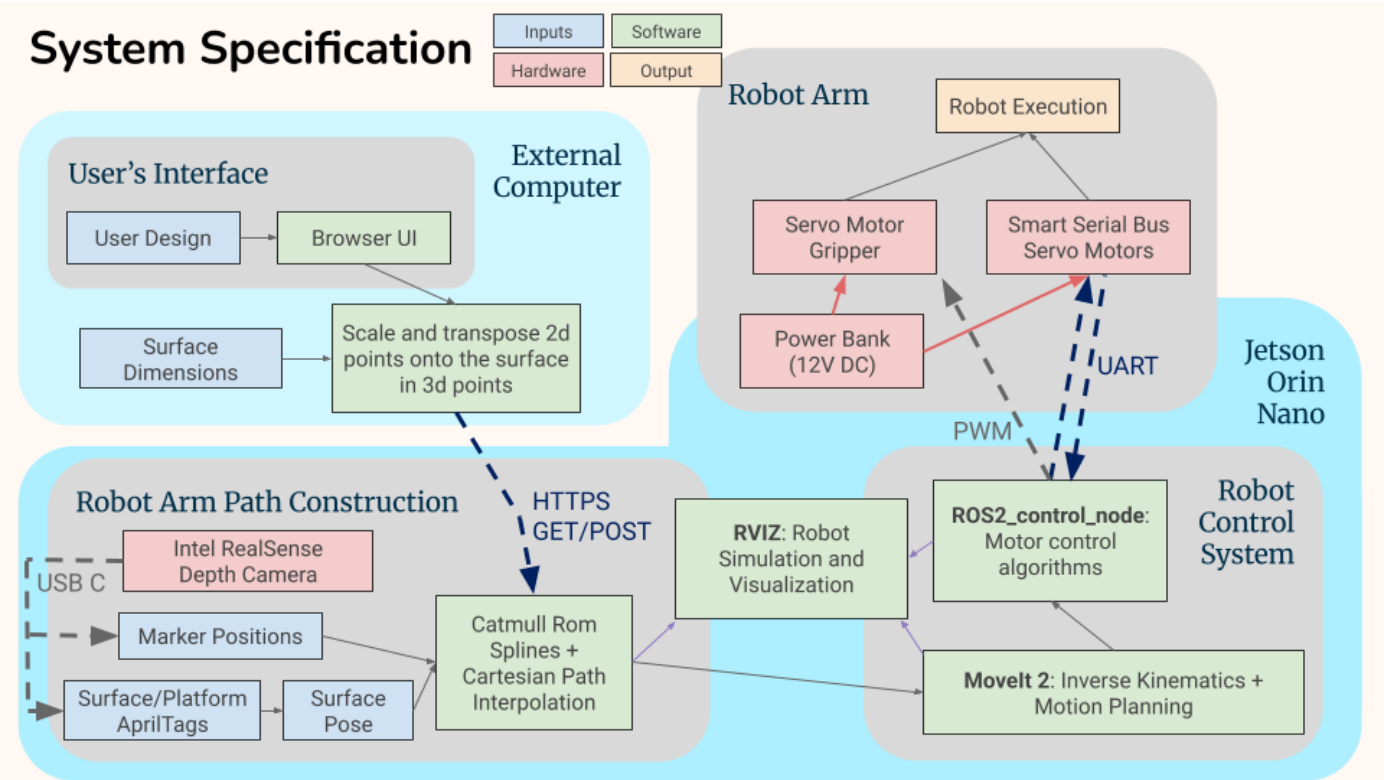


Figure 2: Original system concept. The present design uses platform tags, HTTP, a USB-to-UART driver, manual pen loading, and native SVG sampling. The gripper and Catmull–Rom blocks show later or superseded concepts.

After setup and preview, Start begins the checked drawing while the browser reports progress and offers Stop.

IV. DESIGN REQUIREMENTS

Table I summarizes four engineering specifications derived from Sec. II. The following subsections connect the artist's needs to arm reach, pen orientation, registration, and positioning accuracy.

A. Working Area and Arm Reach

The $0.4 \times 0.4 \times 0.4$ m working area accommodates the object, platform, and pen motion. The separate radial-reach target is at least 0.50 m; the required reach depends on the working area's placement relative to the base. We measure reach from the base-frame origin to the calibrated pen tip:

$$r_{\max} = \max_{q \in \mathcal{Q}} \|p^B_{\text{tip}}(q)\|_2, \tag{1}$$

where q is the joint vector, $\mathcal{Q}$ is the allowed joint range, and p^B_{tip} is the tip position in meters. Link offsets, joint limits, and the table restrict reach in each direction. The chosen placement must permit the full front drawing region and approach and withdrawal motions (Sec. V).

B. Pen Position and Orientation

The six positioning joints let the arm move the pen and keep it perpendicular to the cylinder's front 180-degree drawing region. A round pen can rotate about its own axis without changing the line, giving the planner some freedom to find a smooth motion. Throughout each stroke, the joints must stay within their limits and the arm and holder must clear the object, with only the pen tip touching its surface.

Tool orientation can be described by roll ϕ, pitch θ, and yaw ψ, using the convention

$$R = R_z(\psi)R_y(\theta)R_x(\phi), \qquad R\,\hat{a}^{\text{tool}}_{\text{pen}} = -\hat{n}^B. \tag{2}$$

Here R maps the calibrated pen direction $\hat{a}^{\text{tool}}_{\text{pen}}$ into base coordinates, where it must align with the inward surface normal $-\hat{n}^B$. The rotations R_x, R_y, and R_z describe tool orientation, not individual joints. Rotation about the pen axis remains free to help the planner find smooth motion.

C. Camera Registration Accuracy

The camera locates the platform so the user need not measure its pose manually. The 2 mm target applies to surface points after camera calibration and fixture geometry are included. Since a rotation error $\delta\phi$ shifts a point L meters away by approximately $L\delta\phi$, we check the surface as well as the platform origin and test seating separately.

Table I: Primary engineering specifications.

Specification	Target
Arm positioning freedom	Six revolute joints
Maximum radial reach	≥ 0.50 m to pen tip
Camera registration error	≤ 2 mm position error
Pen-tip positioning error	≤ 2 mm position error

For a known surface point p^O in the object frame, we measure registration error by comparing its estimated and reference positions in the base frame:

$$e_{\text{reg}}(p^O) = \|\hat{p}^B - p^B_{\text{ref}}\|_2. \tag{3}$$

For small pose errors, the point displacement is approximately

$$\delta p^B \approx \delta t + \delta\boldsymbol{\theta} \times (Rp^O), \tag{4}$$

Here δt is translation error, $\delta\boldsymbol{\theta}$ is rotation error in radians, and R rotates object coordinates into base coordinates. Holding fixture geometry fixed gives $e_{\text{reg}} \lesssim \|\delta t\|_2 + L\|\delta\boldsymbol{\theta}\|_2$, so distant points are more sensitive to rotation. We test reseating separately and describe the coordinate conversion in Sec. VI.

D. End-Effector Accuracy

The 2 mm positioning target applies to the calibrated pen tip under operating loads. Calibration, backlash, link bending, and motion tracking affect its accuracy, so encoder feedback alone cannot establish it. The motors must also hold the extended arm with enough torque for acceleration and braking, while sampled paths must preserve the stroke geometry.

Camera and pen-tip accuracy are separate subsystem targets. Their contributions to artwork placement combine with fixture, tool, and path errors:

$$\begin{aligned} e_{\text{draw}} \leq\ & e_{\text{reg}} + e_{\text{fixture}} + e_{\text{arm}} \\ & + e_{\text{tool}} + e_{\text{contact}} + e_{\text{path}}. \end{aligned} \tag{5}$$

These conservative bounds use millimeters and count tool and contact effects separately from e_{arm}. Because errors combine, each 2 mm subsystem target must be tested separately from drawing placement and shape (Sec. VII).

V. DESIGN TRADE STUDIES

A. Arm Configuration and Workspace

We selected six positioning joints to control pen position and orientation along curved strokes, as a planar plotter cannot provide enough control over orientation. Fig. 3 illustrates a 6-DOF arm's workspace.

The SO-101's printed links reduce mechanical design work, but our added joint and pen assembly change mass and inertia. The six-joint URDF and load analysis must include these additions.

Our Python scripts calculate reach and loads from the robot description (URDF), including joint limits, link geometry, masses, centers of mass, and inertias. They include the pen-tip offset, while drawing paths also need orientation and collision checks.

The five-positioning-joint reference model reaches 0.546 m from the base and 0.480 m horizontally. Although the

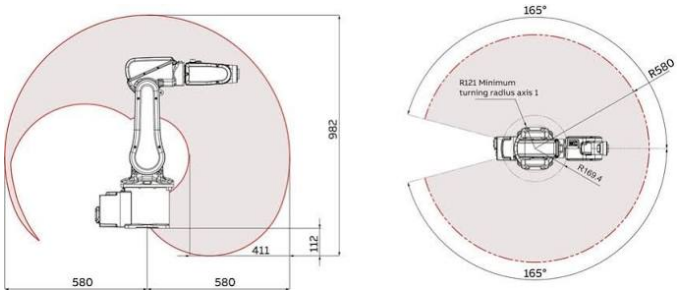


Figure 3: ABB IRB 120 reference workspace [5], with dimensions in millimeters. This illustrates Sec. V.A, not our arm's measured workspace.

first value exceeds 0.50 m, it measures the end frame rather than the pen tip, so the six-joint arm and tool mass still need analysis.

B. Actuator Selection and Static Joint Loads

Smart serial-bus servos provide internal position control and feedback while reducing external driver wiring. Encoder feedback does not measure tool deflection. Each of our six arm actuators has a quoted rating of 30 kgf cm at 12 V, equivalent to

$$\tau_{\text{rated}} = 30(9.80665)(0.01) = 2.942 \text{ N m}. \tag{6}$$

We have not yet confirmed how long the servos can sustain their advertised torque. The usable limit τ_{use} must therefore allow for heating, speed, and duty cycle, with reserve for motion.

Let z_j and o_j be joint j's unit axis and origin in the base frame. For the pen-tip point p, the geometric Jacobian is

$$J_j = \begin{bmatrix} z_j \times (p - o_j) \\ z_j \end{bmatrix}, \quad J = \begin{bmatrix} J_1 & \cdots & J_6 \end{bmatrix}. \tag{7}$$

The upper rows, J_v, relate joint speeds to pen-tip speed, while the lower rows describe rotation. A force F_B at the tip produces joint torque $\tau_F = J_v^T F_B$. Including a moment N_B about the tip gives $J^T W_B$, where $W_B = [F_B^T \ N_B^T]^T$ and both force and moment use base-frame axes.

The gravity-holding moment is calculated independently for each downstream body:

$$g_j(q) = -\sum_{k \in D_j} z_j^T \left[(p_k - o_j) \times m_k \mathbf{g}_B \right]. \tag{8}$$

Here D_j contains downstream bodies, with center of mass p_k, mass m_k in kg, and gravitational acceleration $\mathbf{g}_B$. Each body's lever arm determines its contribution, and the negative sign gives the motor torque that opposes gravity.

Table II reports the reference-model gravity loads found by our scripts. Percentages use the 2.942 N m servo rating rather than the simulator's 3.35 N m limit, and exclude the modified holder, drawing loads, and acceleration.

Table II: Reference-model gravity loads, not final-arm capacity.

Joint	Maximum (N m)	Rating (%)
Shoulder pan	0.000002	< 0.001
Shoulder lift	0.863902	29.36
Elbow flex	0.452765	15.39
Wrist flex	0.117037	3.98
Wrist roll	0.002379	0.08

C. Acceleration and Braking Limits

The horizontal extended-arm pose tests shoulder loading from long gravity lever arms. For upward rotation from rest, holding other joints fixed, a planar approximation gives

$$\begin{aligned} \tau_s &= g_s + I_{\text{eff}} \alpha_s + \tau_{\text{opp}}, \\ \alpha_{s,\max} &= \frac{\tau_{\text{use}} - g_s - \tau_{\text{opp}}}{I_{\text{eff}}}, \\ I_{\text{eff}} &= \sum_k \left(I_{k,\parallel} + m_k r_k^2 \right). \end{aligned} \tag{9}$$

Upward rotation is positive, with gravity compensation g_s, opposing torque τ_{opp}, and acceleration α_s in rad/s^2. Each body's inertia $I_{k,\parallel}$ is measured about a shoulder-parallel center-of-mass axis at shoulder distance r_k. The quoted rating leaves 2.078 N m above baseline gravity load, while acceleration still depends on inertia and usable torque.

Ignoring friction and contact only for illustration, the permissible signed acceleration interval is

$$\frac{-\tau_{\text{use}} - g_s}{I_{\text{eff}}} \le \alpha_s \le \frac{\tau_{\text{use}} - g_s}{I_{\text{eff}}}. \tag{10}$$

Gravity helps brake upward travel but increases the torque needed to brake downward travel. This case is not a universal maximum across all joint motions. For coordinated trajectories, our scripts will evaluate

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau_{\text{act}} + J(q)^T W_{\text{ext}}, \tag{11}$$

where M describes joint inertia and c accounts for forces caused by motion. We must check torque limits in every joint, including those holding position. This follows standard robot dynamics [6], with MuJoCo providing an independent comparison [7].

D. Surface Registration and Fixture Selection

Fixing the target would simplify locating it but restrict the artist's placement. Reconstructing its surface from depth images would allow more shapes at the cost of added geometry processing. We choose a known model on an AprilTag platform to allow repeatable setup without manual pose measurement.

The platform's flat tag-mounting face leaves the artwork clear and avoids mounts shaped for each object. Pegs locate the object or holder, while the seating surface supports its weight. The three identical pegs in Fig. 7 also

require checking which way the object faces. The camera locates the platform, then known fixture geometry locates the object, assuming consistent seating and secure attachment. Platform tags cannot reveal object slip, so calibration and fixture checks remain necessary (Sec. VI, (13)).

E. Artwork Mapping and Path Representation

SVG defines lines, curves, and stroke boundaries directly, while raster artwork first needs tracing. On the cylinder, we map horizontal distance ℓ to arc length through $\theta - \theta_0 = \ell/R$, preserving surface distance as described in Sec. VI.

Perspective projection intersects viewing rays with the surface (Fig. 4), preserving appearance from one viewpoint rather than surface lengths. We choose arc-length mapping for the cylinder.

SVG already defines the artwork's curves [8], so we sample them directly rather than fit another curve through the samples. For a cubic Bezier segment,

$$\begin{aligned}\mathbf{b}(t) = (1-t)^3\mathbf{p}_0 + 3(1-t)^2 t\mathbf{p}_1 \\ + 3(1-t)t^2\mathbf{p}_2 + t^3\mathbf{p}_3,\end{aligned} \tag{12}$$

where $0 \le t \le 1$, $\mathbf{p}_0$ and $\mathbf{p}_3$ are endpoints, and $\mathbf{p}_1$ and $\mathbf{p}_2$ are control points. We sample more closely where the curve needs it, preserving corners and stroke boundaries before mapping onto the surface.

VI. SYSTEM IMPLEMENTATION

This section details the hardware and software that implement the subsystems in Fig. 2.

A. Arm and Pen Holder

Six 12 V smart serial-bus servos drive the arm's joints toward the 0.50 m reach target (Sec. IV). Fig. 5 shows the PLA and PETG assembly (Appendix C), which the final URDF must describe along with the pen-tip offset. Gripping will use a separate servo later.

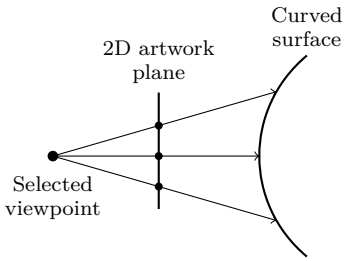


Figure 4: Conceptual cross-section of viewpoint projection onto a curved surface. Dots represent artwork points; rays continue to surface intersections.

The printed sleeve clamps the pen with a thumb screw. We measure the tightened pen's tip offset from the last joint and store it as a fixed URDF transform shared by the planner and Python model. Two C-clamps fix the base to the table.

B. Power and Servo Communication

Fig. 6 shows the wiring. A 12 V DC supply powers the driver board through a normally closed emergency stop. Six servos, IDs 1–6 from base to wrist, share a daisy-chained 12 V, ground, and UART bus. The board connects to the Jetson by USB. Servos accept goal positions and report position, load, voltage, and temperature. The dashed PWM connection shows a later gripping concept; its power and control interface remain to be finalized.

The Jetson has its own adapter, so emergency stop cuts motor power only. Because the unpowered arm sags, software Stop normally holds it in place and emergency stop is a last resort. We will verify the servo feedback against its protocol before using it; reported load is not calibrated torque.

C. Perception and Registration

The base-mounted RealSense D455 sends 1280×720 color images at 30 fps over USB-C [9, 10]. OpenCV [11] detects the platform's tag36h11 AprilTag corners, whose positions are known from CAD. One `solvePnP` calculation uses all visible corners to estimate platform position and orientation, T_P^C. We reject fewer than two visible tags or mean reprojection error above 1 px, which compares observed corners with predicted image positions. Independent surface measurements check the 2 mm target. At depth $Z = 0.5$ m, the nominal 90° horizontal field of view gives $2Z\tan(45°)/1280 \approx 0.78$ mm per pixel; actual distance and camera calibration determine this value. AprilTags therefore let us locate the platform without training an object-recognition model.

With the object or holder seated and its orientation checked (Fig. 7), CAD specifies the fixed object-to-platform pose T_O^P. A surface point $\mathbf{p}_O$ is expressed in the

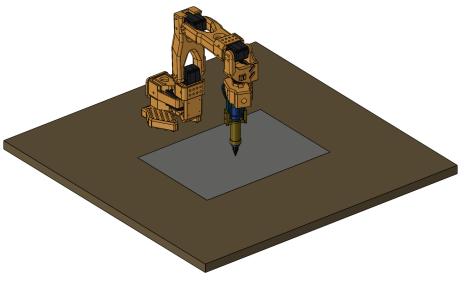


Figure 5: Current robot-arm assembly in CAD, including the pen holder and tabletop mounting.

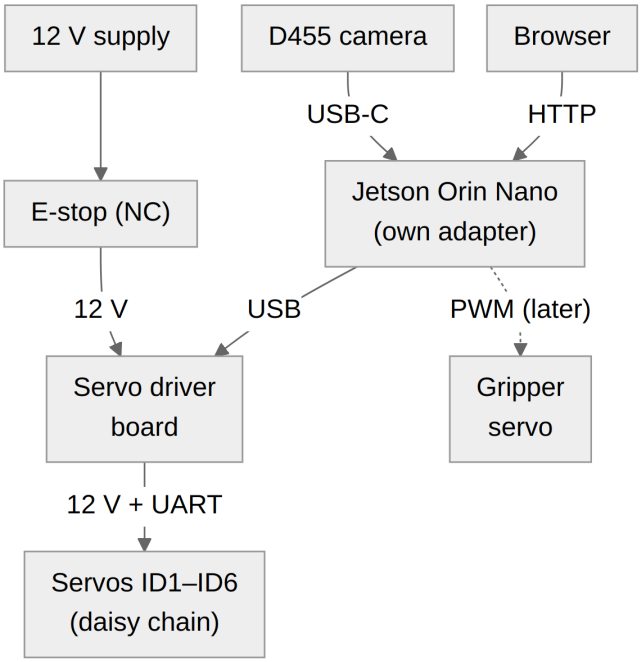


Figure 6: Power and communication wiring. E-stop cuts motor power; the Jetson uses its own adapter. The dashed gripper connection is a later concept.

robot base frame as

$$\tilde{\mathbf{p}}^{B} = T_C^B\, T_P^C\, T_O^P\, \tilde{\mathbf{p}}^{O}, \tag{13}$$

where $\tilde{\mathbf{p}} = [x\ y\ z\ 1]^T$ and each T is a 4×4 homogeneous transform with $3{\times}3$ rotation matrix R and $3{\times}1$ translation vector $\mathbf{t}$:

$$T = \begin{bmatrix} R_{3\times3} & \mathbf{t}_{3\times1} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix}. \tag{14}$$

We calibrate T_C^B by moving the pen tip to four marked platform points and fitting the camera and arm measurements by least squares. At least three points must be non-collinear to determine this transform.

Frame definitions: B is at the robot base; C is at the camera (with z forward and y down); P is halfway between the two tags on the platform's front face; O is at the bottom center of the cylinder with its x-axis pointing toward the robot. Frame O is 102.5 mm along $+x_P$ and 23 mm along $+z_P$ from P, rotated 180° about the platform's z-axis.

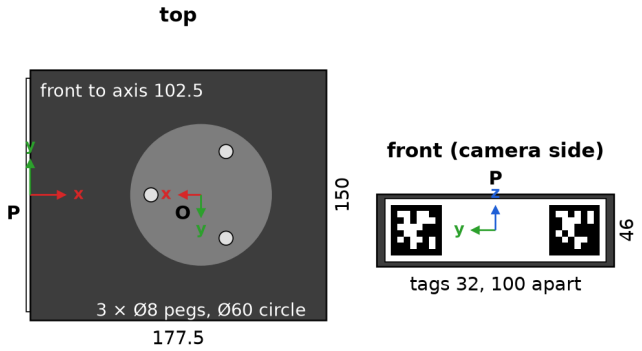


Figure 7: Object-platform top and camera-facing front views, in millimeters. Pegs seat in platform holes; tags occupy the front face.

D. Software Platform and User Interface

The Jetson runs JetPack 6 (Ubuntu 22.04) and ROS2 Humble [12], with components shown in Fig. 8. The tf2 library [13] keeps base, camera, platform, object, and tip coordinates consistent. HTTP GET and POST requests let the browser upload, place, preview, start, and stop artwork jobs. The three checks in Fig. 9 run before motion, while testing will confirm the 50-stroke and 1 m path limits.

E. Artwork Transformations and Mapping

The mapper samples each SVG path's native line and curve geometry directly, using (12) for cubic segments. It preserves stroke boundaries and sharp corners without fitting a new spline through the samples. Each point (u, v) is mapped onto a cylinder of radius R in the object frame:

$$\mathbf{p}_O = \begin{bmatrix} R\cos\theta \\ R\sin\theta \\ z \end{bmatrix}, \quad \theta = \theta_0 + \frac{s\,u}{R}, \quad z = z_0 - s\,v, \tag{15}$$

Here s sets millimeters per SVG unit and (θ_0, z_0) places the drawing. Radius R, height z_0, and mapped positions use millimeters, which we convert to meters for tf2, the URDF, and MoveIt 2. Horizontal distances become surface arc lengths while the pen follows the inward normal $-[\cos\theta\ \sin\theta\ 0]^T$. The limit $|\theta - \theta_{\text{front}}| \le 90°$ keeps points in the front 180 degrees, with θ_{front} checked during setup. Between strokes, the pen lifts outward along the normal by a provisional 10 mm, and we check the full path and holder for clearance.

F. Motion Planning and Servo Control

The planner uses the CAD-based URDF to check joint limits and collisions, then converts stroke points with (13). MoveIt 2 and KDL find the required joint angles [14, 15], starting from the previous angles to help maintain smooth motion. The planner checks sudden joint changes and collisions while allowing rotation about the pen axis, rejecting any job with a stroke it cannot follow continuously.

We assign times to the planned joint movements [16] and send a `FollowJointTrajectory` action to the `ros2_control` controller [17]. Our hardware interface sends servo positions and reads feedback, which the broadcaster publishes on `/joint_states` [18]. The Python model in Sec. V also checks MoveIt's joint angles and their loads.

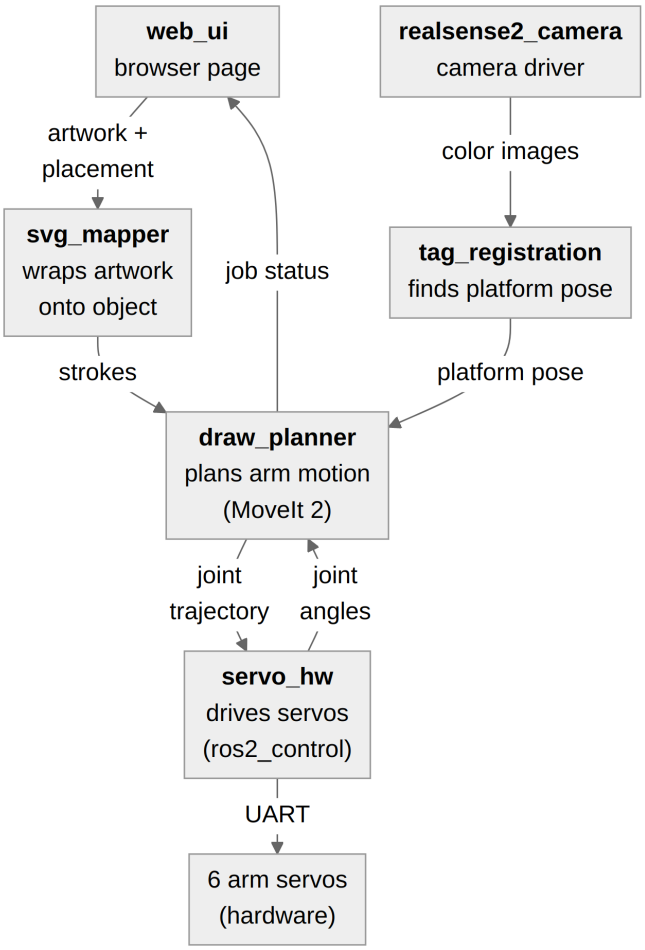


Figure 8: ROS2 nodes and interfaces. Servo commands pass through the USB-to-UART driver board; returned joint angles supply the state interfaces.

VII. TEST, VERIFICATION AND VALIDATION

The tests in this section compare measured errors with the engineering specifications in Sec. IV and the drawing and safety requirements in Sec. II, using (5) to distinguish the error sources.

A. Tests for Registration Accuracy

A ruler-checked 10 mm grid and fixture geometry will establish independent references relative to the arm. Across ten platform placements, including three rotations, estimated surface points must lie within 2 mm of them. We will account for angular displacement $L\delta\phi$ and assess image fit through reprojection error. Ten removals and reseatings will test fixture repeatability, while covering tags must cause rejection when fewer than two remain visible.

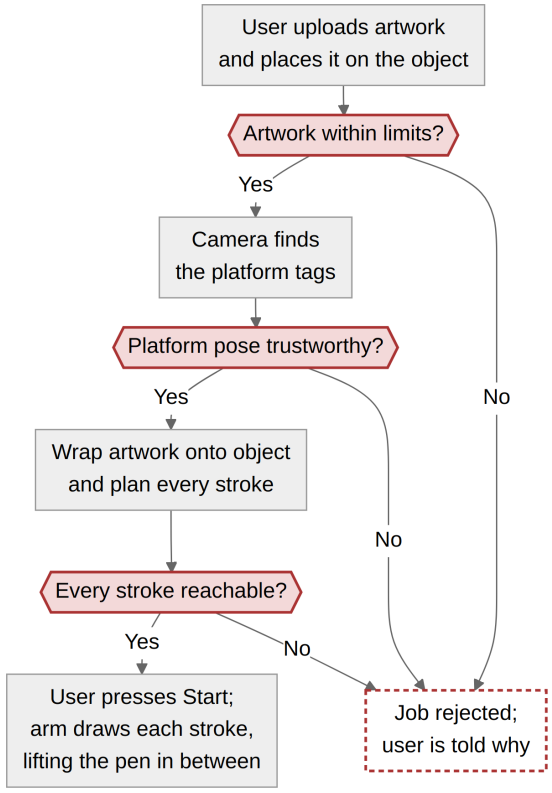


Figure 9: Drawing job flow. A failed check (shaded) rejects the job; Stop holds position at any time.

B. Tests for Arm Accuracy, Reach, and Load

We will measure pen marks with calipers at 20 commanded grid points and three board heights, requiring offsets no greater than 2 mm. An independently measured, collision-free pose must reach 0.50 m from base origin to tip. Five perpendicular pen poses across the front 180 degrees will check reach and record angular error, whose tolerance remains unspecified. We will then hold the most extended permitted pose for ten minutes and test acceleration and braking, requiring no overload or temperature fault and current within the supply rating. Comparing measured torque with (8) requires calibration or a validated conversion; bus load otherwise remains diagnostic.

C. Tests for Drawing Accuracy

We will draw lines, circles, corners, and separate strokes on a paper-wrapped cylinder, then scan the unrolled paper at 600 dpi to compare it with the SVG. Each of five platform placements must reproduce every intended stroke without unintended connections and keep maximum centerline deviation within 2 mm, as required in Sec. II. Translation and rotation may align the scan, but scaling or warping is prohibited. A separate comparison before alignment will measure placement error in independently measured surface coordinates. One 50-stroke, 1 m job will also test the input limits; the shape requirement remains

separate from the registration and pen-tip specifications.

D. Tests for Safe Job Handling

We will test oversized artwork, covered tags, and unreachable strokes (Fig. 9) five times each. All 15 trials must leave the arm stationary and display a reason. At a low pose, ten tests each must confirm software Stop holds position and emergency stop cuts motor power. Later pickup testing requires at least 18 successes in 20 attempts for its provisional 90% goal.

VIII. PROJECT MANAGEMENT

A. Schedule

Fig. 11 in Appendix B schedules three arm iterations from September 13 to December 5, 2026. Iteration 1 is designed and printed, with assembly and wiring due October 17; Iterations 2 and 3 follow by November 7 and 14, each with physical tests. ROS2, inverse kinematics, motion planning, artwork mapping, and the website develop alongside the hardware in simulation. Final verification runs November 8–21, with November 15–December 5 reserved for integration buffer.

B. Team Member Responsibilities

Aaron Tang leads arm CAD, simulation, surface models, artwork mapping, the 2D planner, and website; he supports inverse kinematics and motion planning. Bryan Hong leads assembly, power, servo wiring, emergency stop, and AprilTag detection; he supports the servo hardware interface. Jonathan Waller leads ROS2, workspace and torque analysis, motor tests, inverse kinematics, motion planning, the hardware interface, and tag-to-arm calibration; he supports assembly and wiring. All three share physical testing and integration.

C. Bill of Materials and Budget

Table IV in Appendix C lists the currently recorded components, quantities, costs, and purchase sources. We have spent $230.09 of our $600 budget, leaving $369.91; the Jetson Orin Nano and RealSense D455 camera are borrowed at no project cost.

D. TechSpark Use Plan

We will use TechSpark to laser-cut the acrylic base plate from the extra bin for mounting the arm and platform, after the team member(s) in charge complete laser-cutter training. We will print the other parts on our Bambu Lab A1 in PLA or PETG, with Iteration 1 requiring an estimated 35–40 printing hours and 300 g of filament.

E. Risk Mitigation Plans

Our minimum deliverable is single-color drawing with manual pen loading. Table III links each risk to a test and fallback. Moving the target closer or fixing its position is easier for a demonstration, but it reduces reach and flexible placement, and these fallbacks would not meet all our original specifications.

Table III: Risks, detection, and mitigation actions.

Risk	How we will know	What we will do
Servos cannot hold extended arm	Overload or heat fault (Sec. VII.B)	Move target closer; upgrade shoulder and elbow servos if needed
Camera error above 2 mm	Registration test (Sec. VII.A)	Fix platform; measure its pose with the pen tip instead of camera registration
Pen loses contact	Missing strokes (Sec. VII.C)	Add a spring-loaded pen holder
Fabrication runs late	Iteration 1 incomplete by Oct. 17, 2026	Continue simulation; use buffer from Nov. 15, 2026
Later pickup not ready	Fewer than 18/20 successes (Sec. VII.D)	Retain manual pen loading

IX. RELATED WORK

The EggBot rotates an egg or small sphere under a pen, acting as a spherical-coordinate plotter [19]. This keeps the mechanism simple and inexpensive but requires holding the object on its rotation axis. Our object stays still while the arm draws on its cylindrical region, adding the work of calibrating six joints.

Liu et al. used a UR3 arm for multi-color drawing on cylinders and other 3D surfaces, combining automatic pen pickup, depth-based object location, and force control [20]. We follow their mapping, object-location, and motion-planning sequence, but use platform tags and omit force control to fit the student budget.

Weingartshofer et al. mapped 2D drawings onto complex objects with an industrial robot, comparing motion control with hybrid force/motion control for steady pen contact [21]. Our cylinder mapping in (15) preserves drawing distances as arc length and height, making constant-radius cylinders a simpler starting point.

Our design combines custom mechanics, AprilTag registration, and MoveIt 2 planning while omitting force sensing to fit the student budget. Purchased parts currently total $230.09, with the Jetson and camera borrowed at no cost, so this total excludes their replacement cost.

X. SUMMARY

Leonardo Da Pinchy is designed to let an artist reproduce SVG line art on a known cylinder through a browser. The artist chooses the placement and scale, then the Re-

alSense camera locates the tagged platform so ROS2 on the Jetson can map the strokes and guide the six-joint arm with the pen perpendicular to the surface.

The main challenges are meeting both 2 mm targets, keeping the pen in contact without force sensing, and integrating hardware and software on time. The tests in Sec. VII and milestones in Sec. VIII.A will track progress, while Sec. VIII.E describes fallbacks that preserve a single-color demonstration with reduced capability.

GLOSSARY OF ACRONYMS

CAD – Computer-aided design
DOF – Degrees of freedom
dpi – Dots per inch
fps – Frames per second
HTTP – Hypertext Transfer Protocol
IK – Inverse kinematics
KDL – Kinematics and Dynamics Library
NC – Normally closed
PETG – Polyethylene terephthalate glycol
PLA – Polylactic acid
ROS2 – Robot Operating System 2
SVG – Scalable Vector Graphics
UART – Universal asynchronous receiver/transmitter
UI – User interface
URDF – Unified Robot Description Format
USB – Universal Serial Bus

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APPENDIX A OVERALL SYSTEM DRAWING

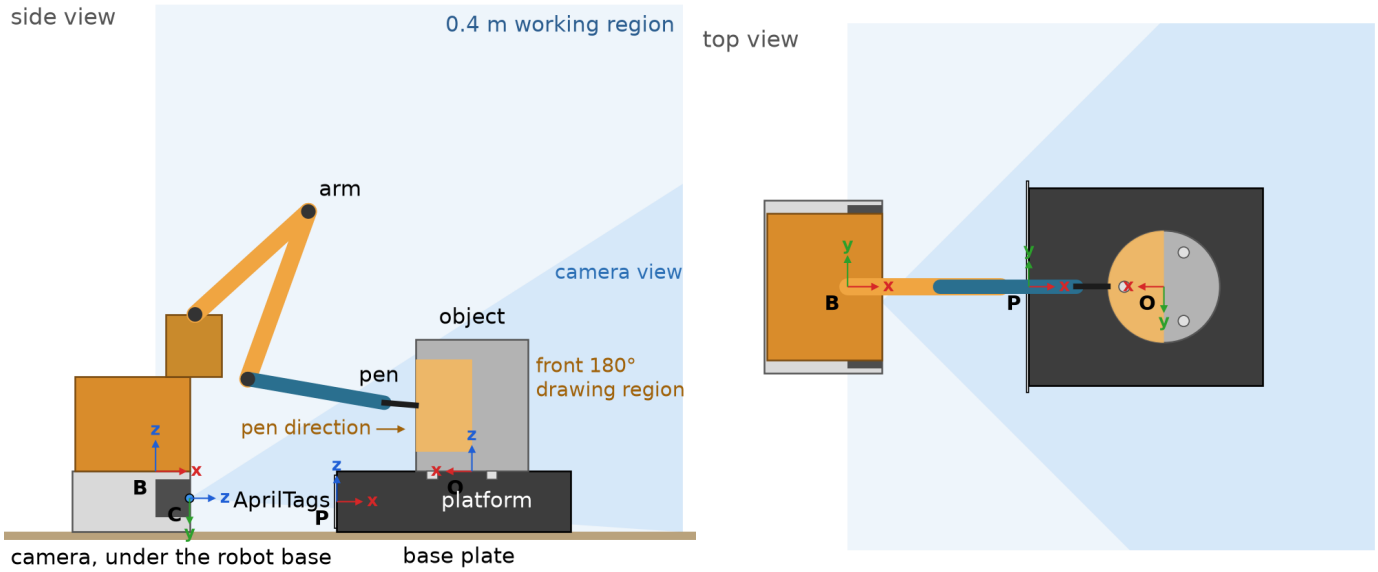


Figure 10: The full physical system view, supplementing Fig. 1 (schematic, not to scale). The working volume is $0.4 \times 0.4 \times 0.4$ m. The D455 under the arm base faces tags on the platform front face. We will print pegs under the object seat in platform holes. The front 180-degree region must be reachable with pen-holder clearance.

APPENDIX B REVISED SCHEDULE AND MILESTONES

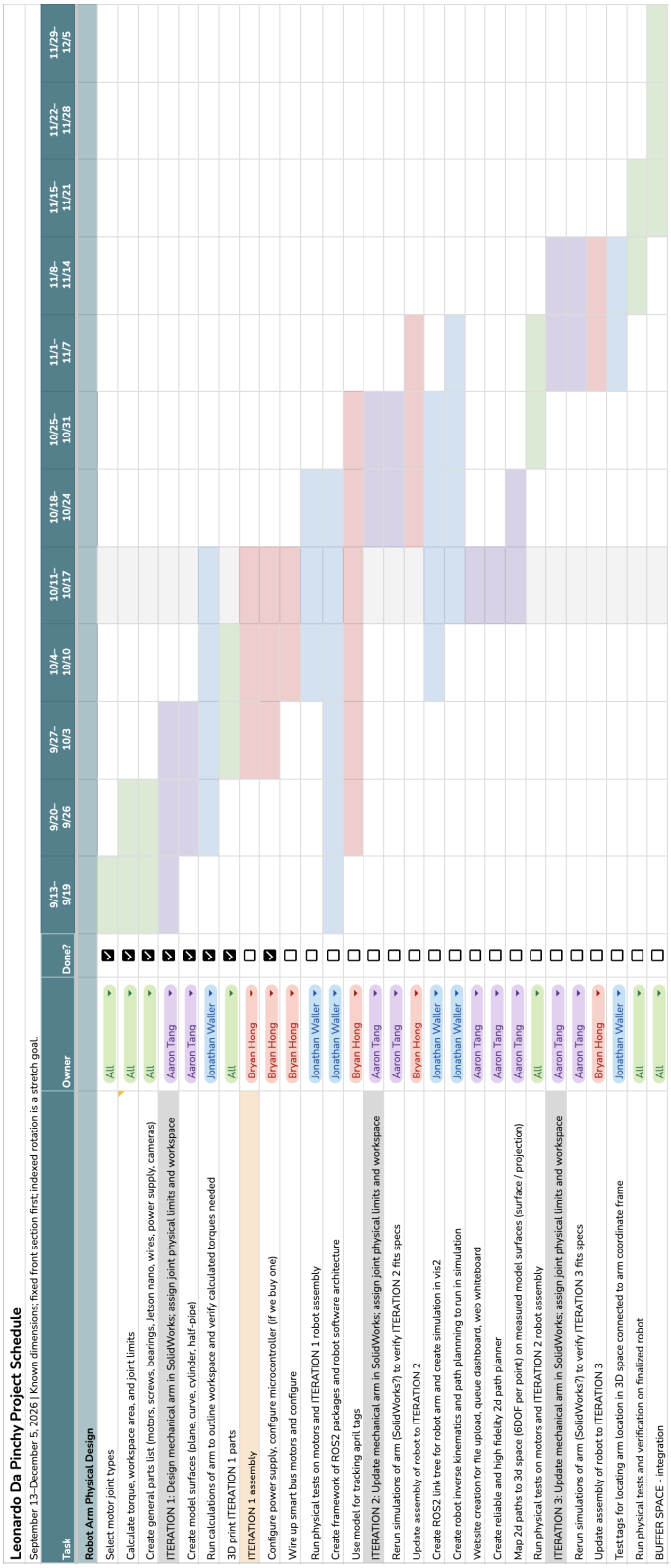


Figure 11: Project schedule and assigned workstreams, September 13–December 5, 2026.

APPENDIX C BILL OF MATERIALS AND BUDGET

Table IV: Bill of materials and budget (items marked “Borrowed” are zero project cost).

Item	Model	Manufacturer	Qty.	Unit cost	Total	Status	Source
Jetson Orin Nano	Variant TBD	NVIDIA	1	$0.00	$0.00	Borrowed	–
Intel RealSense D455	D455	Intel	1	$0.00	$0.00	Borrowed	–
FEETECH serial bus driver board	URT-2	FEETECH	1	$19.99	$19.99	Purchased	https://www.amazon.com/dp/B0GWL4TTR9
RCmall-listed 12V servos (6-pack)	STS3215-12V	FEETECH (sold by RC-mall)	1	$136.28	$136.28	Purchased	https://www.amazon.com/dp/B0FLPQQ4FR
Vise / clamp (2-pack)	Not stated	LEADRISE	1	$10.99	$10.99	Purchased	https://www.amazon.com/dp/B08JYDWMR4
SUNLU PLA filament (1 kg)	SLCA-PLA-BK-1KG-GMSLUS	SUNLUGW (brand SUNLU)	2	$13.99	$27.98	Purchased	https://www.amazon.com/dp/B07XG3RM58
Baomain E-stop	43217-5544	Baomain	1	$9.99	$9.99	Purchased	https://www.amazon.com/dp/B00NTT91Y0
12V power brick	4342705952	SANSUN	1	$9.99	$9.99	Purchased	https://a.co/d/0jhBMIH0
PETG filament (1 kg)	MX-EL-3D-PHE1	ELEGOO	1	$14.87	$14.87	Purchased	https://www.amazon.com/dp/B0CPF7FDTG
	Total project cost (excluding borrowed items)				$230.09		

Six servos will control the six positioning joints; the separate gripping servo would be a later addition and is not included in the purchased total. Borrowed parts are listed at no project cost; the total reflects the purchased components shown.